Streptokinase Karma 1 500 000 IU Powder for Solution for Infusion

Description
White to slight yellow lyophilisate
Each vial contains 1,500,000 IU Streptokinase as a freeze-dried powder for reconstitution.
After reconstitution: clear to slight opalescent, colourless slightly yellow solution.

Pharmacodynamics
Pharmacotherapeutic group: Streptokinase (antithrombotic agents, enzymes)
ATC code: B01A D01

Streptokinase is a highly purified streptokinase derived from β haemolytic streptococci of Lancefield group C. The activation of the endogenous fibrinolytic system is initiated by the formation of a streptokinase-plasminogen complex.

This complex possesses activator properties and converts plasminogen into the proteolytic and fibrinolytic active plasmin. The more plasminogen that is bound within this activator complex, the less plasminogen is left to be converted into its enzymatically active form. Therefore, high doses of streptokinase are associated with a lower bleeding risk and vice versa.

After intravenous administration and neutralisation of the individual antistreptokinase-antibody titre, streptokinase is immediately available systemically for activation of the fibrinolytic system.

Streptokinase has a very short half-life. The first rapid clearance from the plasma is due to the formation of the complex between streptokinase and streptokinase antibody. This complex is biochemically inert and is cleared rapidly from the circulation. Once the antibody has been neutralised, the streptokinase activates the plasminogen as described above.

Pharmacokinetics
The elimination kinetics of streptokinase follows a biphasic course. A small proportion of the dose is bound to antistreptokinase antibodies and metabolised with a half-life of 18 minutes while most of it forms a streptokinase-plaminogen activator complex and is biotransformed with a half-life of about 80 minutes.

Like other proteins, streptokinase is metabolised proteolytically in the liver a nd eliminated via the kidneys.

Indication
Systemic administration

- acute transmural myocardial infarction (not older than 12 hours), with persistent ST-segment elevation or recent left bundle-branch block
- deep vein thromboses (not older than 14 days)
- acute massive pulmonary embolism
- acute and subacute thromboses of peripheral arteries
- chronic occlusive arterial diseases (not older than 6 weeks),
- occlusion of central retinal artery or vein (arterial occlusions not older than 6 to 8 hours, venous occlusions not older than 10 days).

Local administration

- acute myocardial infarction for re-opening of coronary vessels (not older than 12 hours)
- acute, subacute and chronic thromboses as well as embolisms of peripheral venous and arterial vessels.

Note: No statement on therapy outcome can be made for administration beyond the time windows indicated above.

Recommended Dose

Note: When thrombolytic therapy is necessary or a high antibody concentration against streptokinase is present, or recent streptokinase therapy has been given (more than 5 days and less than one year previously), homologous fibrinolytics should be used (see also Special warnings and special precautions for use).

Acute transmural myocardial infarction with persistent ST-segment elevation or recent left bundle-branch block:

Systemic administration

In short-term lysis for the treatment of acute myocardial infarction 1.5 Mio IU Streptokinase are given within 60 min.

Local administration

In acute myocardial infarction patients are given an intracoronary bolus of 20 000 IU Streptokinase on average and a maintenance dose of 2000 IU to 4000 IU per min over 30 to 90 min.

Acute, subacute and chronic thromboses / embolisms of peripheral venous and arterial vessels and chronic occlusive arterial diseases:

Systemic administration

In short-term thrombolysis adults with peripheral venous and arterial vessel occlusions/ embolisms receive an initial dose of 250 000 IU Streptokinase within 30 min, followed by a maintenance dose of 1.5 Mio IU per hour over a maximum of six hours. The six-hour Streptokinase infusion can be repeated on the following day, depending on the therapeutic success of lysis. However, repetition of treatment must on no account be conducted later than 5 days after the first course.

As an alternative to short-term lysis, a long-term lysis for the treatment of peripheral occlusions may be considered. An initial dose of 250 000 IU Streptokinase is given within 30 min, followed by a maintenance dose of 100 000 IU per hour. The duration of therapy depends on the extension and localisation of the vessel occlusion. In peripheral vessel occlusion the maximum duration is 5 days.

Local administration

Patients with acute, subacute and chronic peripheral thromboses and embolisms receive 1000 IU to 2000 IU Streptokinase in intervals of 3 to 5 min. The duration of administration depends on the length and localisation of the vessel occlusion and amounts up to 3 hours at a total dose of max. 120 000 IU Streptokinase.

A percutaneous transluminal angioplasty (mechanical vessel dilatation) can be performed simultaneously, if necessary.

Occlusions of central retinal artery or vein:

Systemic administration
In case of thromboses of the central retinal vessels, lysis of arterial occlusions should be limited to max. 24 hours, in venous occlusions to max. 72 hours. If continuation of thrombolysis is indicated due to extensive thrombotic occlusions, therapy should be interrupted for one day, followed by administration of a homologous fibrinolytic.

Dosage for neonates, infants and children:
Sufficient experience with Streptokinase 1 500 000 IU therapy in children is not yet available. The benefit of treatment has to be evaluated against the potential risks which may aggravate an acute life-threatening situation.

Control of therapy:

Systemic administration
In case of short-term lysis over six hours heparin should be administered during or following Streptokinase infusion when the thrombin time (TT) or partial thromboplastin time (aPTT) have reached less than twice or 1.5 times the normal control value, respectively. The TT and aPTT should be prolonged to 2 to 4 fold and 1.5 to 2.5 fold the normal value, respectively, in order to ensure sufficient protection against rethrombosis (reocclusion of the vessel).

If the Streptokinase infusion is not repeated the heparin therapy is instituted simultaneously with the administration of oral anticoagulants (see Follow-up treatment).

The long-term lysis is controlled with the thrombin time (TT). A 2 to 4 fold prolongation of the TT which is considered as a sufficient anticoagulant protection has to be aimed at. Therefore, a simultaneous administration of heparin may become necessary from the 16th hour of treatment. If the TT after the 16th hour is still prolonged to more than 4 fold the normal control value, the maintenance dose of Streptokinase has to be doubled for several hours until the TT recedes.

Local administration
As is usual with angiographies (x-tray of the vessels with the help of contrast media), heparin is administered - if necessary - prior to the angiography as a safeguard against catheter-induced thromboses. The success of therapy can be determined by the angiography. With a sufficient blood flow of more than 15 minutes the therapy can be considered successful and then terminated.

Follow-up treatment
After every course of streptokinase therapy, a follow-up treatment with anticoagulants or platelet aggregation inhibitors (drugs which inhibit platelet induced clot formation), can be instituted as a prevention of rethromboses. With heparin therapy, in particular, an increased risk of haemorrhage must be considered. The heparin therapy is controlled individually with the TT or aPTT. A 2 to 4 fold prolongation of the TT and 1.5 to 2.5 fold prolongation of the aPTT is aimed at. For long term prophylaxis oral anticoagulants, as - for example - coumarin derivatives platelet aggregation inhibitors can be applied.

Administration
Streptokinase 1500000 IU is administered intravenously and intracoronary. The duration of therapy depends on the nature and extension of the vessel occlusion and differs according to the indication.

Streptokinase 1 500 000 IU is presented as a white to slight yellow lyophilisate. Upon reconstitution with Water for Injection or 0.9% NaCl, clear to slight opalescent, colourless slightly yellow solution obtained.

Route of Administration
Intravenous and Intracoronary

Contraindications
Hypersensitivity to the active substance or to any of the excipients
Contraindications to treatment with Streptokinase, because of the increased risk of haemorrhage under thrombolytic therapy, include:

- existing or recent internal haemorrhage
- all forms of reduced blood coagulability, in particular spontaneous fibrinolysis and extensive clotting disorders
- recent cerebrovascular accident, intracranial or intraspinal surgery
- intracranial neoplasm
- recent head trauma
- arteriovenous malformation or aneurysm
- known neoplasm with risk of haemorrhage
- acute pancreatitis
- uncontrollable hypertension with systolic values over 200 mm Hg and/or diastolic values over 100 mm Hg or hypertensive retinal changes Grades III/IV
- recent implantation of a vessel prosthesis
- simultaneous or recent treatment with oral anticoagulants (INR >1.3)
- severe liver or kidney damage
- endocarditis or pericarditis. Isolated cases of pericarditis, misdiagnosed as acute myocardial infarction and treated with streptokinase, have resulted in pericardial effusions including tamponade
- known haemorrhagic diathesis
- recent major operations (6th to 10th post-operative day, depending on the extent of the procedure)
- invasive operations, e.g. recent organ biopsy, long-term (traumatic) closed chest cardiac massage

Warnings and Precautions
WARNING
Severe bleeding such as intracranial haemorrhage may occur following administration of the drug, particularly in the elderly patients. The risk must be balanced against the potential benefit of thrombolysis.
The following precautions need to be observed:
Patients should be carefully observed for clinical signs during and following administration of the drug for early detection of bleeding. Frequent haematological tests such as blood coagulation tests are mandatory.

To prevent bleeding at the site of centesis or other regions, caution must be exercised concerning procedures and management of arterial/ venus puncture.
The use of heparin in conjunction with the thrombolytic agent for the purpose of prevention of reocclusion may increase the risk of intracranial haemorrhage. Close monitoring of patients is strongly recommended.

The following conditions would normally be considered contraindications to streptokinase therapy, but in certain situations the benefits could outweigh the potential risks:

- recent severe gastrointestinal bleeding, e.g. active peptic ulcer
- risk of severe local haemorrhage, e.g. in case of translumbar aortography
- recent trauma and cardiopulmonary resuscitation
- invasive operations, e.g. recent intubation
- puncture of non-compressible vessels, intramuscular injections, large arteries
- recent abortion or delivery
- pregnancy
- diseases of the urogenital tract with existing or potential sources of bleeding (implanted bladder catheter)
- known septic thrombotic disease
- severe arteriosclerotic vessel degeneration, cerebrovascular diseases
- cavernous pulmonary diseases, e.g. open tuberculosis or severe bronchitis
- mitral valve defects or atrial fibrillation
- aortic dissection
- diabetic retinopathy increase risk of local bleeding

Antistreptokinase
Repeat treatment with streptokinase administered more than 5 days and less than 12 months after initial treatment may not be effective. This is because of the increased likelihood of resistance due to antistreptokinase antibodies.
Also, the therapeutic effect may be reduced in patients with recent streptococcal infections such as streptococcal pharyngitis, acute rheumatic fever and acute glomerulonephritis.

Infusion rate and corticosteroid prophylaxis
At the beginning of therapy, a fall in blood pressure, tachycardia or bradycardia (in individual cases going as far a shock) are commonly observed. Therefore, at the beginning of therapy the infusion should be performed slowly.
Corticosteroids can be administered prophylactically to reduce the likelihood of infusion-related allergic reactions.

Pre-treatment with heparin or coumarin derivatives
If the patient is under active heparinization, it should be neutralised by administering protamine sulphate before the start of the thrombolytic therapy. The thrombin time should not be more than twice the normal control value before thrombolytic therapy is started. In patients previously treated with coumarin derivatives, the INR (International Normalized Ratio) must be less than 1.3 before starting the streptokinase infusion.

Simultaneous treatment with acetylsalicylic acid
Recent evidence indicates that controlled-dose adjuvant acetylsalicyclic therapy in combination with streptokinase is capable of improving the response in the management of

acute myocardial infarction. See also section on Dosing
Streptokinase is not indicated for restoration of patency of intravenous catheters.

Interactions with Other Medicaments
There is an increased risk of haemorrhage in patients who are receiving or who have recently been treated with anticoagulants, e.g. heparin or drugs which inhibit platelet formation or function, e.g. platelet aggregation inhibitors, dextrans.

Pregnancy and Lactation
Streptokinase is contraindicated in pregnancy. There is no evidence of the drug's safety in pregnancy, nor is there evidence from animal work that it is free from hazard. Bleeding and anaphylactic reactions might cause abortion and foetal death, especially when streptokinase is given within the first 18 weeks of pregnancy. Use only when there is no safer alternative. It is unknown whether streptokinase is excreted in human milk. Breast milk should be discarded during the first 24 hours following thrombolytic therapy.

Side Effects
Blood and lymphatic system disorders
Common: haemorrhage at the injection site, ecchymoses, gastrointestinal bleeding, genitourinary bleeding, epistaxis
Uncommon: cerebral haemorrhages with their complications and possible fatal outcome, retinal haemorrhages, severe haemorrhages (also with fatal outcome), liver haemorrhages, retroperitoneal bleeding, bleeding into joints, splenic rupture. Blood transfusions are rarely required.
Very rare: haemorrhage into the pericardium including myocardial rupture during thrombolytic treatment of acute myocardial infarction

In serious haemorrhagic complications, streptokinase therapy should be discontinued and a proteinase inhibitor, e.g., aprotinin, should be given as follows. Initially 500 000 KIU (Kallikrein Inactivator Unit) up to one million KIU by slow intravenous injection or infusion. If necessary this should be followed by 200,000 KIU every four hours by intravenous drip until the bleeding stops. In addition, combination with synthetic antifibrinolytics is recommended. If necessary, clotting factors can be substituted. Additional administration of synthetic antifibrinolytics has been reported to be efficient in single cases of bleeding episodes.

Immune system disorders
Very Common: development of antistreptokinase antibodies
Common: allergic anaphylactic reactions, e.g. rash, flushing, itching, urticaria, angioneurotic oedema, dyspnoea, bronchospasm, hypotension
Very Rare: delayed allergic reactions, e.g. serum sickness, arthritis, vasculitis, nephritis, neuroallergic symptoms (polyneuropathy, e.g. Guillain Barré syndrome), severe allergic reactions up to shock including respiratory arrest.
Allergic reactions can largely be avoided by giving the infusion slowly. Moderate or mild allergic reactions can be managed with concomitant antihistamine and/or corticosteroid therapy. If a severe allergic reaction occurs the infusion of streptokinase should be discontinued immediately and the patient given the appropriate treatment. The current medical standards for shock treatment should be observed. Lysis therapy should be continued with homologous fibrinolytics, such as Urokinase or tPA.

Nervous system disorders
Rare: neurologic symptoms (e.g. dizziness, confusion, paralysis, hemiparesis, agitation, convulsion) in the context of cerebral haemorrhages or cardiovascular disorders with hypoperfusion of the brain

Eye disorders
Very rare: iritis/uveitis/iridocyclitis

Cardiac and vascular disorders
Common: at the start of therapy, hypotension, tachycardia, bradycardia
Very rare: crystal cholesterol embolism

During fibrinolytic therapy with streptokinase in patients with myocardial infarction, the following events have been reported as complications of myocardial infarction and/or symptoms of reperfusion:
Very common: hypotension, heart rate and rhythm disorders, angina pectoris
Common: recurrent ischaemia, heart failure, reinfarction, cardiogenic shock, pericarditis, pulmonary oedema
Uncommon: cardiac arrest (leading to respiratory arrest), mitral insufficiency, pericardial effusion, cardiac tamponade, myocardial rupture, pulmonary or peripheral embolism

These cardiovascular complications can be life-threatening and may lead to death.
During local lysis of peripheral arteries, distal embolization cannot be excluded.

Respiratory Disorders
Very rare: non-cardiogenic pulmonary oedema after intracoronary thrombolytic therapy in patients with extensive myocardial infarction

Gastrointestinal disorders
Common: nausea, diarrhoea, epigastric pain, vomiting

General disorders and administration site conditions
Common: headache, back pain, musculoskeletal pain, chills, fever, asthenia, malaise

Testing
Common: Transient elevations of serum transaminases and bilirubin

Symptoms and Treatment of Overdose
Long-term overdosage of streptokinase may induce the risk of rethrombosis by prolonged decrease of plasminogen. See also sections on Side Effects and Pharmacodynamic properties.

Effects on Ability to Drive and Use Machines
Not Applicable.

Preclinical Safety Data
Not Applicable.

Instructions for Use
The contents should be dissolved in 5 ml of Water for Injection or 0.9% NaCl. The solution should be swirled gently to facilitate quick reconstitution, but care should be taken to avoid foaming.

Streptokinase Karma may be given by intravenous infusion in 50-200 ml of 5% glucose solution.

Any unused medicinal product or waste material should be disposed of in accordance with local requirements.

This product is compatible with 5% glucose.

Incompatibilities
This medicinal product must not be mixed with other medicinal products.

Storage Condition
Store below 30°C. Do not freeze the product. The reconstituted solution should not be stored for more than 24 hours in a refrigerator at 2°C to 8°C.

Shelf Life
The shelf-life of unopened vials of Streptokinase Karma stored below 30°C is 36 months. Do not freeze the product. The reconstituted solution should not be stored for more than 24 hours in a refrigerator at 2°C to 8°C.

Presentation
Streptokinase Karma is supplied in 8R vial, 8.00 ml hydrolytic, colourless glass, Ph.Eur. type I, with rubber closures (bromobutyl polymer) and aluminium seal with plastic flip-top caps (red color). The secondary packaging format is 34 x 34 x 64 mm and the material is 300 g/m² GT2. Streptokinase Karma is available in packages containing one vial.

Manufactured by:
Lyocontract GmbH
Pulverwiese 1, Trift, Ilsenburg, D-38871, Germany

For:
Karma Pharmatech GmbH
Emil-von-Behring-Straβe 76,
35041 Marburg/Lahn, Germany.

Revision date
10/10/2021